

Li Ju

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Education

Ph.D., Scientific Computing Uppsala University, Uppsala, Sweden	Sep. 2021 – Jun. 2026
M.Sc., Computational Science Uppsala University, Uppsala, Sweden	Sep. 2019 – Aug. 2021
M.Sc., Chemometrics, <i>with Honors</i> University of Science and Technology of China, Hefei, China	Sep. 2016 – Jun. 2019
B.Sc., Chemistry University of Science and Technology of China, Hefei, China	Sep. 2012 – Jun. 2016

Professional Experience

Doctoral Researcher Scientific Machine Learning Laboratory, Uppsala University Advisors: Andreas Hellander, Prashant Singh	Sep. 2021 –
Machine Learning Engineer Intern Modulai, Stockholm, Sweden	Jan. 2026 – Apr. 2026
Data Scientist (Part-time) Scaleout Systems AB, Uppsala, Sweden	Jan. 2021 – Jun. 2021

Research Interests

- Probabilistic and generative modeling
- Uncertainty quantification for multimodal foundation models
- Post-hoc adaptation and representation geometry of pre-trained encoders
- Distributed and federated learning algorithms and systems

Selected Publications

Full publication record available on [Google Scholar](#).

Conference

Ju, L., Nautiyal, M., Vats, E., & Singh, P. (2026). Epistemic Uncertainty Quantification for Pre-trained VLMs via Riemannian Flow Matching. *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*.

Ju, L., Andersson, M., Fredriksson, S., Glöckner, E., Hellander, A., Vats, E., & Singh, P. (2025). Exploiting the Asymmetric Uncertainty Structure of Pre-trained VLMs on the Unit Hypersphere. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*.

Ju, L., Singh, P., & Toor, S. (2021). Proactive Autoscaling for Edge Computing Systems with Kubernetes. In *Proceedings of the 14th IEEE/ACM International Conference on Utility and Cloud Computing Companion (UCC 2021)*.

Journal

Ju, L., Zhang, T., Toor, S., & Hellander, A. (2024). Accelerating Fair Federated Learning: Adaptive Federated Adam. *IEEE Transactions on Machine Learning in Communications and Networking*, **2**, 1017–1032.

Ju, L., Hellander, A., & Spjuth, O. (2024). Federated Learning for Predicting Compound Mechanism of Action Based on Image Data From Cell Painting. *Artificial Intelligence in the Life Sciences*, **5**.

Under Review

Nautiyal, M., **Ju, L.,** Hellander, A., Vats, E., & Singh, P. (2026). GeoFlowVLM: Geometry-Aware Joint Uncertainty for Frozen Vision-Language Embeddings. *arXiv:2605.13352*.

Nautiyal, M., **Ju, L.,** Ernfors, M., Hagland, K., Holma, V., Söderholm, M. W., Hellander, A., & Singh, P. (2026). OneFlowSBI: One Model, Many Queries for Simulation-Based Inference. *arXiv:2601.22951*.

Honors and Awards

1st Place in Europe, 2nd Place Globally Huawei Wireless Communication Global Hackathon	2025
2nd Place, Huawei Sweden Hackathon Huawei Sweden	2024
M.Sc. in Chemometrics, <i>with Honors</i> University of Science and Technology of China	2016 – 2019
Second-Class Freshman Scholarship University of Science and Technology of China	2012

Invited Talks & Presentations

Invited Talks

Sep. 2025	“Orthogonality in Neural Networks,” Scaleout Edges, Sweden.
May 2024	“Federated Learning for Predicting Compound Mechanism of Action Based on Image Data from Cell Painting,” AI4Research, Uppsala University.
Apr. 2024	“Denoising Diffusion Probabilistic Models,” Ph.D. Course on Bayesian Inference, Uppsala University.
Dec. 2023	“Federated Learning from the Perspective of Optimization,” Centre for Interdisciplinary Mathematics, Uppsala University.

Posters

Oct. 2024	“Is Logit Adjustment a Free Lunch for Heterogeneous Federated Learning?” eSSENCE 2024, Sweden.
Oct. 2022	“Accelerating Fair Federated Learning,” eSSENCE 2022, Sweden.

Teaching

Head Teaching Assistant Data Engineering II (1TD076), Uppsala University	2026
Teaching Assistant Data Engineering II (1TD076), Uppsala University	2022 – 2025
Guest Lecturer Distributed and Parallel Computing (1TD070), Uppsala University	2025
Teaching Assistant Data Engineering I (1TD169), Uppsala University	2021 – 2022
Supervisor Supervised four master thesis/course projects on uncertainty quantification, generative modeling, and simulation-based inference, Uppsala University.	2023 – 2026

Service

Leadership

President of Uppsala Chapter of the Society for Industrial and Applied Mathematics 2024 – 2025
Uppsala University, Uppsala

Academic Reviewing

ICML 2026, NeurIPS 2026, IEEE TNNLS, IEEE TMC, IEEE CM.

Open-Source Software

Blades (Contributor) 2022 – 2023
Simulator and benchmark suite for Byzantine-robust federated learning.

FEDn (Contributor) 2020 – 2021
Open-source federated learning framework; contributed PyTorch integration.

Technical Skills

Frameworks	PyTorch, JAX, transformers, verl, vllm, FastAPI, pthread, OpenMP, MPI, CUDA
Cloud / DevOps	Docker, Apptainer, Kubernetes, OpenStack
Programming Languages	Python, C/C++, Haskell, Erlang, Lisp/Racket
Spoken Languages	Chinese (native), English (fluent)

References

Assoc. Prof. Andreas Hellander Department of Information Technology, Uppsala University andreas.hellander@it.uu.se	Ph.D. advisor
Assoc. Prof. Prashant Singh Science for Life Laboratory & Department of Information Technology, Uppsala University prashant.singh@it.uu.se	Co-advisor
Assis. Prof. Ekta Vats Department of Information Technology, Uppsala University ekta.vats@it.uu.se	Collaborator

Last updated: June 8, 2026